Round-Number Anchoring in Overnight US Equity Order Sizes: Evidence from 274 Million Order Events on the Blue Ocean ATS

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Abstract

Order sizes on Blue Ocean ATS, the dominant overnight trading venue for US National Market System stocks, exhibit pronounced clustering at round-number anchors. Across 273.9 million add events on 30 stocks between August 2025 and March 2026, 40.0 percent of orders sit at exactly one of 34 round-number share counts. The single most common size is 20 shares (5.27 percent of all adds), with secondary peaks at 1 share (4.67 percent), 100 shares (2.84 percent), and 200 shares (2.83 percent). The 20-share mode is anomalous relative to the round-lot conventions documented for NYSE and NASDAQ in prior work (Alexander and Peterson 2007), where modes occur at 500, 1000, and 5000 shares. Hartigan dip tests reject unimodality at the 5 percent level for 30 of 31 tickers; Gaussian mixture model selection by BIC prefers nine or ten components with means clustered at round-number share counts and several components fitting essentially as point masses at exact integers. Cross-ticker round-number clustering ranges from 15 percent (TQQQ) to 67 percent (DIA), and the size of the modal anchor scales with ticker price level. The findings extend the size-clustering literature to a new venue with structurally distinct characteristics — single-venue continuous overnight market dominated by sponsored-access retail flow — and document a sub-100-share modal anchor that has no counterpart in the published literature on US regular-hours equity venues.

Keywords Order-size clustering · Round-number heuristics · Overnight trading · Blue Ocean ATS · Market microstructure · Retail trading · Behavioral finance

JEL Classification G14 · G15 · G23 · D87

1 Introduction

Order sizes in equity markets exhibit clustering on round numbers. Trades on NYSE and NASDAQ concentrate at multiples of 500, 1,000, and 5,000 shares (Alexander and Peterson 2007). Size clustering is identified as a stylised fact of limit order books across markets, alongside price clustering, persistence in order signs, and high cancellation rates. The economic explanations have a long pedigree: round-number anchoring as a cognitive heuristic (Wyckoff 1963; Niederhoffer and Osborne 1966), as a negotiation device that reduces search costs (Harris 1991), and as a stealth-trading tactic that disguises informed flow within retail-style order patterns (Barclay and Warner 1993; Alexander and Peterson 2007).

This paper documents the size-clustering pattern on Blue Ocean ATS, a venue with three structurally distinctive features that the existing literature has not addressed. First, Blue Ocean is a single-venue continuous market: from 8:00 PM to 4:00 AM Eastern Time, no other lit US equity venue operates, eliminating the competitive multi-venue setting that characterises the studies in the established literature. Second, order flow is dominated by retail accounts, principally Asia-Pacific clients accessing the venue through US broker-dealer subscribers (Korea Times 2026). Third, the venue is new in kind: institutionally accessible high-resolution market data became available in August 2025, opening a window that prior work could not have studied.

I use full message-by-order data from Databento (dataset identifier OCEA.MEMOIR) on 30 actively traded US securities from August 2025 to March 2026, comprising 273.9 million order add events. The data permit identification of order placements at the level of individual orders

rather than aggregated trade prints. I test for multimodality in the order-size distribution and characterise the round-number anchors empirically through Gaussian mixture models in log-space. Three findings emerge. First, multimodality in order sizes is universal: 30 of 31 tickers reject unimodality under the Hartigan dip test at the 5 percent level, and Gaussian mixture model selection by BIC prefers nine or ten components for every ticker. Several mixture components fit essentially as point masses at exact integer share counts, indicating that the multimodality reflects literal round-number anchors rather than smooth heavy-tailed regions of mass. Second, the round-number share is high: 40.0 percent of pooled order placements sit at one of 34 round-number sizes (1, 2, 3, 4, 5, 10, 15, 20, 25, 30, 40, 50, 60, 75, 100, 125, 150, 200, 250, 300, 400, 500, 600, 700, 750, 800, 900, 1000, 1500, 2000, 2500, 3000, 5000, 10000). Third, the modal anchor is anomalous: 5.27 percent of all order placements are at exactly 20 shares, the single largest mode in the data. This contrasts with the existing literature on NYSE and NASDAQ (Alexander and Peterson 2007), which identifies modes at 500, 1,000, and 5,000 shares but not at sub-100-share sizes.

Cross-ticker variation in clustering intensity is substantial. The round-number share ranges from 15.5 percent (TQQQ, a leveraged ETF) to 67.3 percent (DIA, an index ETF). The location of the modal anchor scales with ticker price level: high-priced tickers (META, MSFT) cluster at 20 shares; medium-priced tickers (AAPL) cluster at 40 shares; low-priced ETFs (TQQQ, IWM) cluster at higher absolute share counts.

The contribution is descriptive and modest in scope. The paper extends the trade-size clustering literature (Alexander and Peterson 2007; ap Gwilym and Meng 2010; Verousis and ap Gwilym 2013; Lin et al. 2019; Chen 2019) to a structurally distinctive venue and documents a non-standard modal anchor that the literature does not predict. The 20-share anomaly is reported but not explained beyond noting that, given its persistence across the entire eight-hour BOATS session, it

most plausibly reflects a structural feature of the venue's order entry — sponsoring broker default sizes, fee-tier optimisation, or platform UI conventions — rather than a temporal-flow phenomenon.

Two companion papers using the same dataset address distinct empirical questions: Lim (2026, SSRN) examines overnight execution-cost premia between Blue Ocean and NASDAQ regular trading hours via the Huang-Stoll spread decomposition; a separate working paper measures price discovery against CME futures during Asia hours via Hasbrouck information shares. The present paper is methodological and descriptive in focus, sharing only the underlying dataset with those works, and is orthogonal to their empirical questions.

The remainder of the paper proceeds as follows. Section 2 reviews the institutional setting and prior literature. Section 3 describes the data. Section 4 describes the multimodality and clustering measures. Section 5 reports the results. Section 6 discusses interpretation and limitations. Section 7 concludes.

2 Institutional setting and related literature

2.1 Blue Ocean ATS

Blue Ocean ATS is a US-registered alternative trading system operating an electronic limit order book for National Market System stocks. The session runs from 8:00 PM to 4:00 AM Eastern Time on Sunday through Thursday evenings, overlapping Asia-Pacific business hours. Order types are limited to day limit orders. Short selling is disabled. The matching engine applies a 20 percent price band against a reference price set at 7:30 PM Eastern, rejecting orders priced beyond that band. There is no minimum order size requirement; orders of any positive integer share count are accepted. No other US equity venue operates a continuous lit order book during this overnight window during the sample period.

Direct subscriber access is restricted to US broker-dealers; non-US institutional and retail accounts access the venue through sponsored connectivity. Korean retail flow dominates the order base (Korea Times 2026). The Blue Ocean overnight session is the single largest source of US equity price discovery during Asia trading hours.

2.2 Prior literature on order-size clustering

Order- and trade-size clustering has been documented across multiple market settings. Alexander and Peterson (2007) document trade-size clustering on NYSE and NASDAQ at multiples of 500, 1,000, and 5,000 shares, and link the pattern to the stealth trading hypothesis (Barclay and Warner 1993): informed traders disguise large trades within the round-size flow generated by uninformed traders. Moulton (2005) finds size clustering in the foreign exchange market at round notional amounts. ap Gwilym and Meng (2010) document size clustering in the FTSE100 index futures market, identifying a trade-off between size resolution and price resolution. Verousis and ap Gwilym (2013) extend the analysis to the London Stock Exchange. Chen (2019) examines the relationship between trade-size clustering and price efficiency. Lin, Chen, and Lien (2019) examine joint price and size clustering on the Taiwan Stock Exchange and document differential clustering across investor types. Related work on order-splitting behaviour (Chou and Wang 2009 on the Taiwan Futures Exchange) and on stealth trading in emerging markets (Cai et al. 2006 on China) provides complementary perspectives on how trader type and venue characteristics interact with order-size choice.

The existing literature establishes three findings of relevance to the present paper. First, size clustering is heavily concentrated at multiples of 100 shares and above; the documented modal sizes are 500, 1000, and 5000. Second, size clustering varies systematically across investor types and across market structures: institutional traders and order-driven markets exhibit different

clustering patterns than retail-dominated and quote-driven settings. Third, theoretical mechanisms for size clustering — behavioural anchoring, negotiation cost reduction, and stealth trading — are not mutually exclusive and operate jointly.

Three features distinguish the Blue Ocean venue from the settings studied in this literature. First, the venue is single-venue and continuous during its session, eliminating the competitive multi-venue dynamics studied for NYSE/NASDAQ comparisons. Second, order flow is dominated by retail accounts accessed through international sponsored connectivity, rather than the mixed institutional/retail flows examined for major US venues. Third, message-by-order data permit observation of order placement decisions at the level of individual orders, rather than aggregated trade-level data used in much of the prior work. The combination of these features motivates a fresh empirical examination.

3 Data

I obtain Blue Ocean ATS message-by-order data through Databento under the dataset identifier OCEA.MEMOIR. The schema records every order event — additions, cancellations, modifications, trades, and fills — with nanosecond timestamps and a persistent order identifier that links the lifecycle of each order. The data structure permits direct identification of order placement decisions, distinct from execution events.

The sample comprises 30 actively traded US securities: five exchange-traded funds (SPY, QQQ, DIA, IWM, TQQQ) and 25 single-name stocks (AAPL, AMD, AMZN, AVGO, BA, BAC, COIN, CSCO, DIS, GOOGL, JPM, KO, META, MSFT, MU, NFLX, NVDA, PFE, PLTR, SMCI, T, TSLA, VZ, WMT, XOM). The single-name set spans large-cap technology, communication services, consumer discretionary, consumer staples, financials, healthcare, industrials, and energy sectors. The sample period is 24 August 2025 to 30 March 2026, spanning 240 ticker-month files.

Across this universe and window the archive contains 273,908,859 add events, the unit of analysis throughout this paper.

The analysis is restricted to order add events (action code A in the schema), not cancels, modifications, or trades. Add events represent the explicit choice by a market participant to place a new limit order at a specified price and size. Restricting to adds isolates the order-placement decision from subsequent management actions, and from the realised execution flow, which can have different size distributions than the underlying order flow.

4 Methodology

4.1 Multimodality test

I assess multimodality of the per-ticker and pooled order-size distributions using two complementary criteria. The first is the Hartigan dip test against the null hypothesis of unimodality. The dip statistic measures the maximum deviation between the empirical cumulative distribution function and the closest unimodal cumulative distribution function. I implement an approximation that fits a piecewise-linear unimodal envelope through quantiles and computes the maximum deviation, with a bootstrap p-value based on samples drawn from the unimodal envelope. The test is run on log-transformed sizes to compress the heavy right tail.

The second criterion is Bayesian Information Criterion (BIC) selection over Gaussian mixture models in log-space with k from one to ten components. For each candidate k, I fit a univariate Gaussian mixture with random initialisation, two restarts, and up to 200 iterations. The mixture model that minimises BIC is selected. To make the fits computationally tractable, I subsample to 200,000 observations per ticker; for tickers with fewer observations the full sample is used.

4.2 Atom verification

Several Gaussian components in the BIC-preferred mixtures have fitted standard deviations approaching zero. This suggests that the components are fitting essentially point masses at exact integer share counts rather than diffuse log-normal regions. To verify directly, I compute the empirical mass at each integer share count from 1 to 100, and at selected higher round numbers (125, 150, 200, 250, 300, 400, 500, 600, 700, 750, 800, 900, 1000, 1500, 2000, 2500, 3000, 5000, 7500, 10000).

The atom verification provides a direct empirical check on the mixture model's interpretation. If sizes have substantial mass at exact integer values relative to neighbouring integers, the mixture's near-zero variance components are picking up real point masses rather than smooth modes.

4.3 Round-number share

I define a round-number set R as the union of {1, 2, 3, 4, 5, 10, 15, 20, 25, 30, 40, 50, 60, 75} (small round numbers) with {100, 125, 150, 200, 250, 300, 400, 500, 600, 700, 750, 800, 900, 1000, 1500, 2000, 2500, 3000, 5000, 10000} (larger round numbers). The choice of R is motivated by inspection of the BIC-selected mixture component means and the empirical atom verification, both of which identify mass concentrated at these values. The round-number share for ticker i is the fraction of add events with size in R.

5 Results

5.1 Universal multimodality

Hartigan dip tests reject the null hypothesis of unimodality at the 5 percent level for 30 of 31 tickers (30 individual tickers plus the pooled sample). Only JPM fails to reject ($p = 1.000$), reflecting the conservative behaviour of the bootstrap approximation under thin samples (1.2

million adds, the smallest among single names). The pooled distribution rejects unimodality decisively ($p < 0.001$ with 200 bootstrap replications).

BIC selection over $k = 1$ to 10 components prefers $k \geq 9$ for every tested distribution. The pooled sample prefers $k = 10$; AAPL, AMD, AMZN, AVGO, COIN, DIS, GOOGL, IWM, META, MSFT, NFLX, PLTR, QQQ, SPY, T, TSLA, VZ, WMT, XOM, BA, KO, all index ETFs, and several other tickers prefer $k = 10$. The remaining tickers (TQQQ, TSLA, MU, NVDA, BAC, CSCO, JPM, PFE, SMCI) prefer $k = 9$. No ticker prefers $k = 1$; no ticker prefers $k < 3$. The data are decisively multimodal in the BIC sense.

The component variances in the fitted mixtures are uniformly small. For the pooled $k = 10$ fit, four of the ten components have fitted log-space standard deviation below 0.05 (i.e., the 95 percent fitted spread covers less than ±10 percent around the component mean in shares). Several components have fitted standard deviation reported as 0.00 by the GMM, consistent with point-mass behaviour at exact integer share counts.

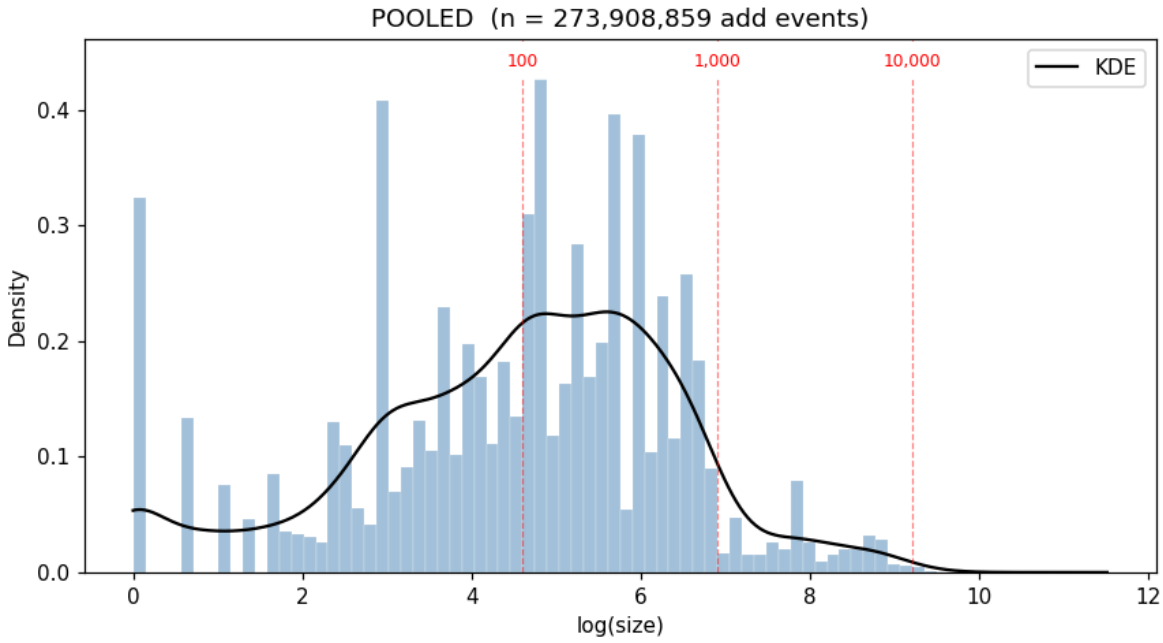


Figure 1 Pooled distribution of log(order size) across 273.9 million add events on 30 stocks, August 2025 to March 2026. Bars show the empirical histogram (binned on log-shares); the black line is a Gaussian kernel density estimate (bandwidth = 0.2). Vertical red dashed lines mark log(100), log(1000), and log(10000). The histogram displays sharp spikes at log(20) ≈ 3.0, log(100) ≈ 4.6, and at intermediate round-number values, indicating multimodality concentrated at integer share-count anchors.

5.2 Round-number atoms

Direct verification confirms that the small-variance components correspond to literal point masses at integer share counts. Table 1 reports the most common single-share atoms in the 1 to 100 range, ordered by frequency, together with selected higher round numbers. The top mode is 20 shares, occupying 5.27 percent of all 273.9 million add events. The second mode is 1 share at 4.67 percent. The next four modes are 100 shares (2.84 percent), 40 shares (1.96 percent), 2 shares (1.92 percent), and 10 shares (1.56 percent). Among higher round numbers, 100, 200, 300, 400, 500, 700, and 800 shares each carry between 1 and 3 percent of pooled mass; multiples of 1000 above 1000 itself are smaller (1000 at 0.64 percent, 2000 at 0.13 percent, 5000 at 0.06 percent).

The point-mass interpretation is decisive. At 20 shares specifically, 14.4 million orders sit at exactly 20 shares, against approximately 1.2 million at 19 shares and 1.0 million at 21 shares (not shown in Table 1). The empirical mass concentrates at the integer 20, not in a Gaussian neighbourhood around it. The pattern repeats at 100, 200, 500, 1000 shares: each integer carries an order of magnitude more mass than its immediate neighbours.

The 5.27 percent mass at 20 shares is the most striking single feature of the distribution and is anomalous relative to the prior literature. Alexander and Peterson (2007) document NYSE and NASDAQ trade-size modes at 500, 1000, and 5000 shares. ap Gwilym and Meng (2010) document size clustering in FTSE100 index futures, identifying a trade-off between size and price resolution. Verousis and ap Gwilym (2013) report comparable patterns on the London Stock Exchange. None

of these venues exhibits a sub-100-share mode of comparable magnitude. The 20-share anchor on Blue Ocean is a venue-specific feature that the existing literature does not predict.

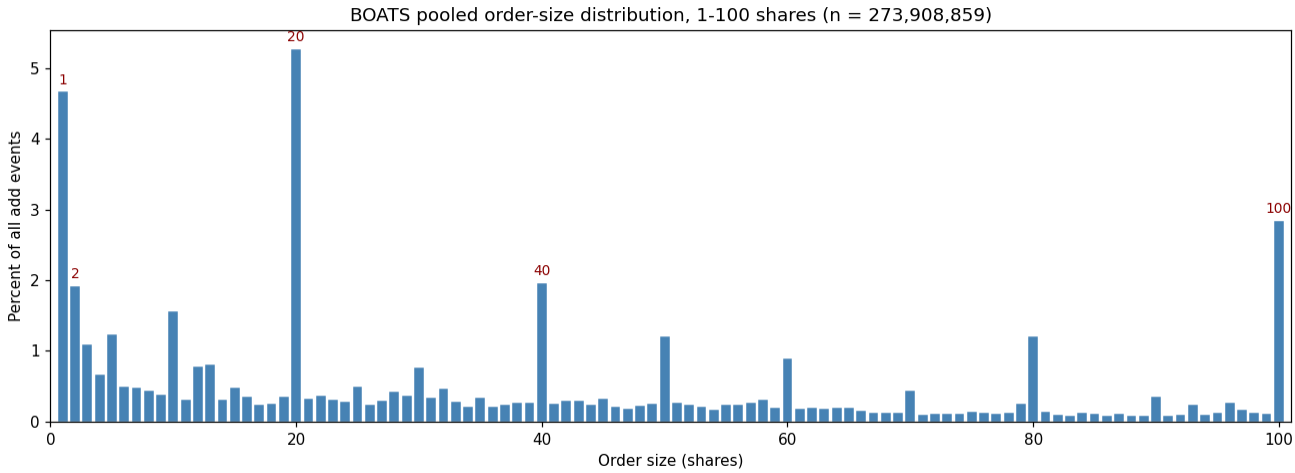


Figure 2 Pooled order-size distribution across the 1 to 100 share range, in percent of all 273.9 million add events. The 20-share mode is the global maximum at 5.27 percent. Secondary modes appear at 1, 100, 40, 2, and 10 shares. Smaller modes are visible at multiples of 10 (10, 30, 40, 50, 60, 70, 80, 90, 100). The distribution is dominated by integer point masses rather than smooth heavy-tailed regions. Top-five atoms in the displayed range are annotated.

Table 1 Empirical mass at single-share atoms (1-100, top 20) and at selected higher round numbers, pooled across 30 tickers and 7 months. The top mode is 20 shares (5.27 percent of all adds), an order of magnitude larger than any single non-round neighbouring integer.

Order size (shares)	Number of add events	Share of pooled adds
Single-share atoms (1-100), top 20		
20	14,438,249	5.27%
1	12,784,425	4.67%
100	7,780,990	2.84%
40	5,362,661	1.96%
2	5,258,178	1.92%
10	4,274,281	1.56%
5	3,367,510	1.23%
50	3,313,165	1.21%
80	3,285,055	1.20%

Order size (shares)	Number of add events	Share of pooled adds
3	2,977,923	1.09%
60	2,462,395	0.90%
13	2,213,621	0.81%
12	2,133,301	0.78%
30	2,078,962	0.76%
4	1,816,809	0.66%
6	1,375,589	0.50%
25	1,339,073	0.49%
15	1,332,673	0.49%
7	1,312,409	0.48%
32	1,279,651	0.47%
Selected higher round numbers		
200	7,738,166	2.83%
300	5,618,294	2.05%
400	6,510,188	2.38%
500	6,536,062	2.39%
700	3,924,744	1.43%
800	4,948,645	1.81%
1000	1,739,013	0.64%
2000	354,966	0.13%
5000	153,219	0.06%
10000	42,446	0.02%

5.3 Cross-ticker heterogeneity

Table 2 reports the per-ticker round-number share alongside the share at four specific anchors (1, 100, 500, and 1000 shares). Round-number share varies substantially across tickers, from 15.5 percent (TQQQ) to 67.3 percent (DIA). Index ETFs are at the higher end of the range (DIA 67, IWM 53, SPY 55, QQQ 47, TQQQ 15) but TQQQ is the lowest in the entire sample. Single names

span a wide range, from BA (65) and NFLX (59) at the high end to TSLA (26), AVGO (24), and AMD (23) at the low end.

The location of the modal anchor varies systematically with ticker price level. High-priced single names (META at $647, MSFT at $413, NVDA at $183) cluster heavily at the 20-share atom, where the dollar value of an order is approximately the same order of magnitude as a low-priced ETF round lot. Medium-priced single names (AAPL at $262, COIN at $212) cluster more diffusely between 20 and 100 shares. ETFs at lower price levels (TQQQ at $50, IWM at $252) cluster at higher absolute share counts (median TQQQ order size is 1,300 shares, IWM 460 shares). The pattern is consistent with a dollar-value anchoring interpretation: market participants approximate a target dollar exposure and round the share count to the nearest convenient anchor at the relevant scale.

Differential mass at specific anchors is also informative. T (AT&T) shows 23.9 percent of all adds at exactly 500 shares — by far the largest single-anchor concentration in the sample. VZ (Verizon) shows 19.4 percent at 500 shares. BAC and PFE show 13 and 12 percent respectively. These four tickers are all dividend-paying, low-priced single names with substantial retail income-investor flow; the 500-share anchor likely reflects the standard size used by structured income-investing accounts.

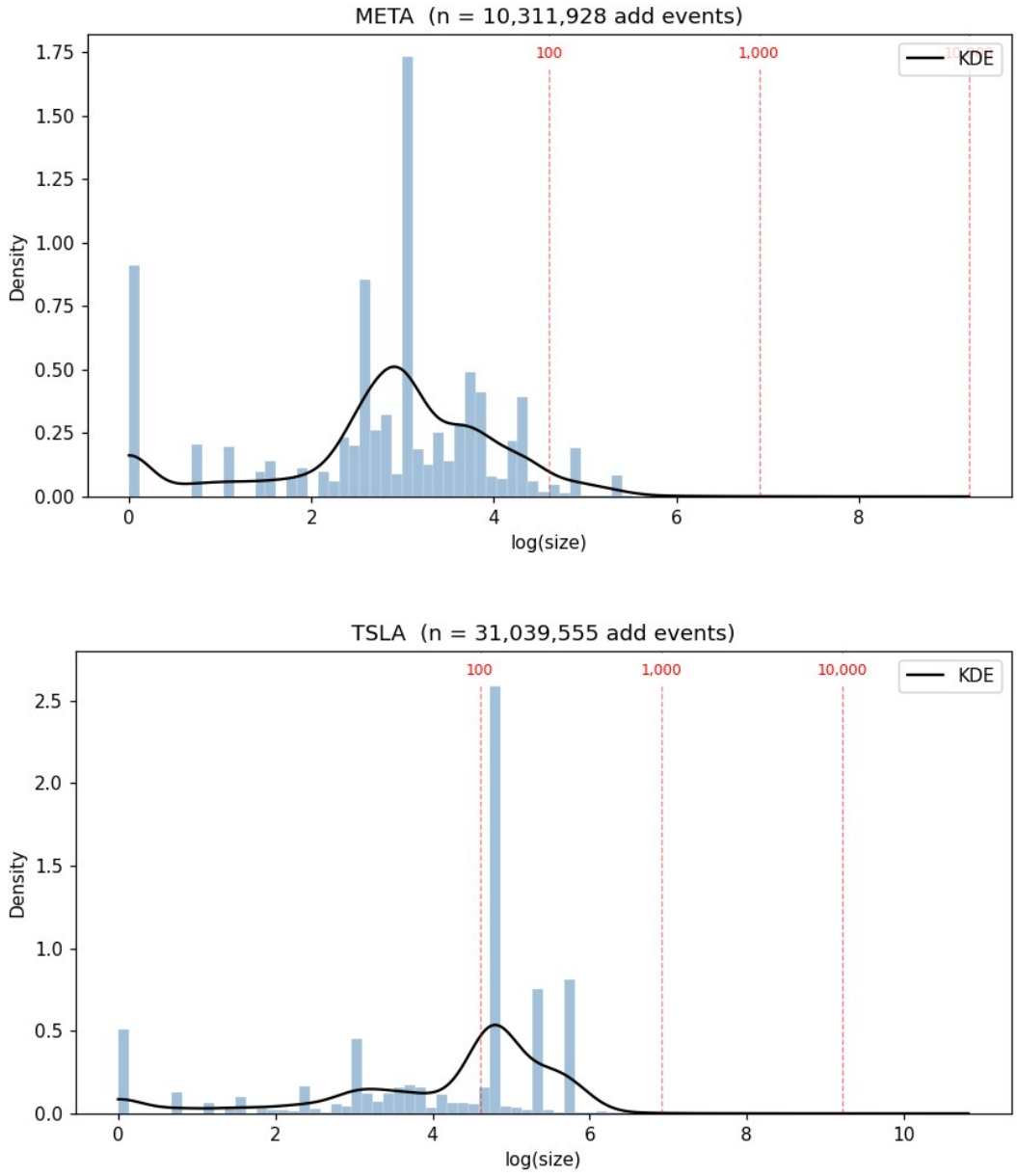


Figure 3 Cross-ticker variation in modal anchor location. Top: META (high-priced single name, n = 10.3 million adds) clusters sharply at log(20) ≈ 3.0 with very little mass beyond log(100). Bottom: TSLA (medium-priced single name, n = 31.0 million adds) clusters dramatically at log(119) ≈ 4.78, with the 119-share anchor accounting for approximately 35 percent of GMM component weight. Both tickers reject unimodality and prefer ten and nine mixture components respectively, but the location of the dominant anchor differs substantially. The pattern is consistent with dollar-value anchoring: META trades around \$647 per share while TSLA trades around \$364 per share during the sample period.

Table 2 Per-ticker round-number share and selected anchor masses. Round share is the fraction of add events with size in the 34-element round-number list defined in Section 4.3. Columns @1,

@100, @500, @1000 give the share at each specific anchor. Tickers within each category are ordered by descending round share.

Ticker	Category	N adds	Round %	@1	@100	@500	@1000
DIA	Index ETF	9,761,103	67.3%	0.4%	5.5%	30.3%	3.6%
IWM	Index ETF	13,190,536	53.0%	0.3%	6.6%	10.0%	8.6%
SPY	Index ETF	18,170,596	54.5%	1.2%	3.8%	1.8%	0.0%
QQQ	Index ETF	39,520,424	47.0%	1.4%	0.3%	0.6%	0.0%
TQQQ	Index ETF	24,843,842	15.5%	1.9%	0.8%	2.1%	0.6%
BA	Single name	978,221	64.7%	9.5%	1.8%	0.1%	0.1%
NFLX	Single name	3,396,776	58.5%	10.5%	11.1%	0.2%	0.1%
GOOGL	Single name	9,684,966	55.1%	8.3%	14.3%	0.5%	0.0%
MSFT	Single name	7,544,491	54.5%	10.8%	1.3%	0.0%	0.0%
T	Single name	511,470	53.4%	8.0%	1.7%	23.9%	0.4%
PLTR	Single name	9,236,867	52.1%	7.6%	2.7%	0.4%	0.1%
DIS	Single name	713,216	51.2%	9.8%	2.1%	0.4%	0.0%
META	Single name	10,311,928	51.1%	10.5%	0.5%	0.0%	0.0%
VZ	Single name	657,944	50.9%	7.3%	3.9%	19.4%	0.2%
AMZN	Single name	7,131,158	48.1%	7.4%	5.0%	0.4%	0.0%
PFE	Single name	1,489,020	44.9%	6.8%	1.6%	12.4%	0.6%
KO	Single name	828,985	43.8%	13.9%	1.4%	2.9%	0.1%
AAPL	Single name	5,361,626	43.4%	9.1%	4.9%	0.9%	0.0%
JPM	Single name	1,197,425	43.1%	8.7%	1.9%	0.0%	0.0%
WMT	Single name	770,423	39.6%	11.7%	1.6%	1.0%	0.1%
COIN	Single name	10,565,678	39.2%	7.1%	4.0%	0.1%	0.0%
MU	Single name	12,830,377	37.8%	2.7%	1.8%	0.1%	0.0%
NVDA	Single name	18,377,948	38.1%	7.7%	2.9%	0.8%	0.1%
XOM	Single name	1,058,347	38.1%	6.2%	4.8%	0.9%	0.1%
BAC	Single name	751,547	36.7%	5.1%	2.1%	13.0%	0.5%
SMCI	Single name	4,228,390	34.5%	2.4%	10.4%	4.1%	0.2%
TSLA	Single name	31,039,555	25.9%	6.8%	1.2%	0.1%	0.0%
CSCO	Single name	711,453	24.7%	3.1%	1.5%	4.3%	0.1%

Ticker	Category	N adds	Round %	@1	@100	@500	@1000
AVGO	Single name	15,427,501	23.9%	4.4%	0.5%	0.0%	0.0%
AMD	Single name	13,617,046	22.5%	3.5%	2.0%	0.1%	0.0%
Pooled	**All**	**273,908,859**	**40.0%**	**4.7%**	**2.8%**	**2.4%**	**0.6%**

5.4 Mixture component structure

Table 3 reports the BIC-preferred mixture component means and weights for the pooled distribution and five illustrative tickers. The pooled fit identifies modes at 1, 2, 10, 20, 35, 60, 111, 264, 644, and 3137 shares; the modal weight (24.5 percent) is on the 264-share component, which captures the cluster of mass at 200, 250, and 300 shares. AAPL identifies modes at 1, 2, 5, 10, 14, 20, 40, 95, 188, and 353 shares with the largest weight on the 95-share component (24.7 percent), capturing the 100-share round lot. TSLA peaks at 119 shares with a 35 percent component weight; this captures a venue-specific TSLA convention not visible at other tickers. TQQQ is the unusual case: with median order size of 1,300 shares, its modes are at 1, 3, 12, 60, 181, 613, 1329, 2609, and 5744 shares, mostly large and highly variable, with weights distributed across the upper components. PFE places 36.5 percent of weight on a 972-share component, near the 1000-share anchor.

Two patterns are notable. First, every ticker's mixture identifies the 1-share atom as a distinct component, with weight ranging from 1.9 percent (TQQQ) to 10.4 percent (KO). The 1-share order is universal across the sample. Second, the larger round-number anchors (100, 500, 1000) appear as separate components for tickers where their mass is non-negligible, but are absorbed into the heavy right tail for tickers with extreme price levels (TQQQ) or where the round-lot convention is shifted by ticker price (TSLA at 119 shares rather than 100).

Table 3 BIC-preferred Gaussian mixture models for pooled distribution and selected tickers. Component means are reported in shares (sorted ascending). Component weights are aligned with means. k* is the number of components selected by BIC over the range k = 1 to 10.

Ticker	k*	Component means (shares, sorted)	Component weights (sorted by mean)
Pooled (n = 274M)	**10**	**1, 2, 10, 20, 35, 60, 111, 264, 644, 3137**	**0.05, 0.02, 0.11, 0.05, 0.09, 0.09, 0.14, 0.25, 0.15, 0.05**
AAPL	10	1, 2, 5, 10, 14, 20, 40, 95, 188, 353	0.09, 0.04, 0.06, 0.03, 0.05, 0.09, 0.24, 0.25, 0.09, 0.08
TSLA	9	1, 2, 7, 21, 36, 83, 119, 217, 307	0.07, 0.02, 0.08, 0.06, 0.07, 0.16, 0.35, 0.09, 0.11
TQQQ	9	1, 3, 12, 60, 181, 613, 1329, 2609, 5744	0.02, 0.02, 0.04, 0.04, 0.12, 0.18, 0.16, 0.20, 0.22
META	10	1, 2, 3, 7, 14, 20, 37, 76, 142, 444	0.10, 0.02, 0.02, 0.09, 0.20, 0.19, 0.27, 0.07, 0.04, 0.00
PFE	9	1, 2, 7, 22, 45, 246, 400, 500, 972	0.07, 0.02, 0.08, 0.07, 0.02, 0.18, 0.07, 0.12, 0.37

6 Discussion

6.1 The 20-share anomaly

The dominant 20-share mode is the most distinctive empirical feature of the Blue Ocean order distribution. It is anomalous in three senses. First, it is well below the 100-share round-lot convention that historically governs US equity markets. Second, no published study of US, European, or Asia-Pacific equity venues has identified a 20-share mode of this magnitude. Third, the mode is robust across single-name tickers with widely different price levels, suggesting that the anchor is a venue-level structural feature rather than a stock-specific phenomenon.

Three explanations are plausible but cannot be distinguished with the available data. First, sponsoring broker-dealers may set 20-share defaults in the order entry interfaces presented to non-US clients. Korean retail brokerages, which dominate Blue Ocean access (Korea Times 2026), serve clients accustomed to Korean Stock Exchange order sizes that operate under different lot conventions. The 20-share default may be a localised UI choice that propagates through to a substantial fraction of order placements. Second, the 20-share size may be optimised against Blue Ocean's fee schedule or related structural costs that this paper does not analyse. Third, the 20-share anchor may reflect a behavioural convention specific to Asia-Pacific retail traders engaging with

US equity markets — a hypothesis that would require survey or experimental evidence to test directly.

I do not attempt to identify the cause. The contribution of this section is to document the empirical anomaly precisely and to flag that the 100-share round-lot assumption embedded in much of the US microstructure literature does not hold on this venue.

6.2 Implications for microstructure measurement

The atomic structure of order sizes has direct implications for measurement procedures that assume continuous size distributions. Liquidity measures based on average order size, simple percentile cutoffs, or parametric size models will be sensitive to the round-number anchors. Methodologies that classify order types as small, medium, or large based on a priori thresholds (e.g., the 100-share round lot as the institutional minimum) will misclassify a substantial fraction of activity on retail-dominated venues like Blue Ocean. Researchers studying overnight US equity microstructure should anchor any size-based filter on empirically grounded boundaries derived from the actual size distribution rather than on conventions imported from regular-hours studies.

The Gaussian mixture decomposition reported in Table 3 provides one approach to such empirical anchoring. The component means identify natural boundaries between cluster groups, and the component weights provide a principled basis for measures conditioned on order-size group membership.

6.3 Limitations

Several limitations bound the interpretation of the findings. First, the data do not contain broker-dealer or sponsored-access client identifiers; I cannot directly attribute the 20-share mode to specific brokers or to specific subsets of retail flow. Second, the analysis covers add events only, not subsequent cancellations, modifications, or trades. Different conditioning would produce

different distributions; in particular, executed trade-size distributions on Blue Ocean may be substantially different from order-add distributions. Third, the sample is restricted to 30 actively traded large-capitalisation US securities; smaller stocks may exhibit different anchor patterns. Fourth, the findings cover a single venue during its monopoly phase; the patterns may shift once exchange overnight sessions become operational.

Two methodological caveats also apply. The Hartigan dip test approximation used here is conservative, and its bootstrap p-values are stable but not exact under the venue's heavy-tailed structure. The Gaussian mixture BIC selection caps at k = 10; the true optimal k may be higher, and the components reported should be interpreted as a parsimonious decomposition of the round-number anchor structure rather than as a fully saturated description of the distribution.

7 Conclusion

Order sizes on Blue Ocean ATS exhibit pronounced round-number clustering. Across 273.9 million order add events on 30 stocks between August 2025 and March 2026, 40 percent of orders sit at one of 34 round-number sizes, with the dominant mode at exactly 20 shares (5.27 percent of all adds). The Hartigan dip test rejects unimodality for 30 of 31 tested distributions, and BIC-selected Gaussian mixtures prefer nine or ten components per ticker, with several components fitting essentially as point masses at integer share counts. Cross-ticker round-number share ranges from 15 percent to 67 percent.

The findings extend the trade- and order-size clustering literature (Alexander and Peterson 2007; ap Gwilym and Meng 2010; Verousis and ap Gwilym 2013; Lin et al. 2019) to a structurally distinctive venue: a single-venue continuous overnight market dominated by sponsored-access retail flow from Asia-Pacific accounts. The 20-share modal anchor is anomalous relative to the

round-lot conventions documented for US regular-hours venues and merits specific attention from researchers studying overnight US equity microstructure.

Future work could pursue three extensions. First, a comparison of size clustering on Blue Ocean against NASDAQ or NYSE during regular hours, on the same securities, would directly characterise how the round-lot anchor structure differs between overnight retail-dominated and daytime competitive venues. Second, identification of the mechanism behind the 20-share anomaly — sponsoring broker UI defaults, fee-schedule optimisation, or behavioural anchoring among Asia-Pacific retail clients — would require ground-truth data on order origination not available in the public data feed. Third, an analysis of how size clustering changes once national securities exchanges activate overnight sessions and the Blue Ocean monopoly fragments would track whether the venue-specific anchors persist or dissolve under competitive pressure.

Declarations

Availability of data and materials

Blue Ocean ATS message-by-order data (Databento dataset identifier OCEA.MEMOIR) were obtained under commercial licence from Databento and cannot be redistributed under the terms of the licence. The Python analysis pipeline (panel construction, multimodality testing, atom verification, and Gaussian mixture fitting) is available from the author upon reasonable request. Aggregated summary tables underlying Tables 1 through 3 are available from the author upon reasonable request. Researchers wishing to replicate the results can obtain equivalent source data directly from Databento under commercial licence.

Competing interests

The author declares no competing interests. The author is an independent researcher with no affiliation to Blue Ocean Technologies, Databento, or any of the exchanges or alternative trading systems discussed in this paper. The author does not hold any financial position that would materially benefit from the findings or interpretations presented here.

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Declaration of generative AI use in scientific writing

The author used Anthropic Claude as a writing aid during the preparation of this manuscript. All empirical results, methodology, and interpretations are the author's responsibility.

References

Alexander GJ, Peterson MA (2007) An analysis of trade-size clustering and its relation to stealth trading. Journal of Financial Economics 84(2):435–471

ap Gwilym O, Meng L (2010) Size clustering in the FTSE100 index futures market. Journal of Futures Markets 30(5):432–443

Aşçıoğlu A, Comerton-Forde C, McInish TH (2007) Price clustering on the Tokyo Stock Exchange. The Financial Review 42(2):289–301

Ball CA, Torous WN, Tschoegl AE (1985) The degree of price resolution: The case of the gold market. Journal of Futures Markets 5(1):29–43

Barclay MJ, Warner JB (1993) Stealth trading and volatility: Which trades move prices? Journal of Financial Economics 34(3):281–305

Bessembinder H (1994) Bid-ask spreads in the interbank foreign exchange markets. Journal of Financial Economics 35(3):317–348

Blau BM, Van Ness BF, Van Ness RA (2012) Trade size and price clustering: The case of short sales and the suspension of price tests. Journal of Financial Research 35(2):159–182

Boehmer E, Jones CM, Zhang X, Zhang X (2021) Tracking retail investor activity. Journal of Finance 76(5):2249–2305

Cai BM, Cai CX, Keasey K (2006) Which trades move prices in emerging markets? Evidence from China's stock market. Pacific-Basin Finance Journal 14(5):453–466

Chen T (2019) Trade-size clustering and price efficiency. Japan and the World Economy 49:195–203

Chou RK, Wang YY (2009) Strategic order splitting, order choice, and aggressiveness: Evidence from the Taiwan futures exchange. Journal of Futures Markets 29(12):1102–1129

Christie WG, Schultz PH (1994) Why do NASDAQ market makers avoid odd-eighth quotes? Journal of Finance 49(5):1813–1840

Hansen PR, Lunde A (2006) Realized variance and market microstructure noise. Journal of Business & Economic Statistics 24(2):127–161

Harris L (1991) Stock price clustering and discreteness. Review of Financial Studies 4(3):389–415

Ikenberry DL, Weston JP (2008) Clustering in US stock prices after decimalisation. European Financial Management 14(1):30–54

Johnson H (2014) Odd lot trades: The behavior, characteristics, and information content over time. The Financial Review 49(4):669–684

Kandel S, Sarig O, Wohl A (2001) Do investors prefer round stock prices? Evidence from Israeli IPO auctions. Journal of Banking & Finance 25(8):1543–1551

Korea Times (2026) Blue Ocean Technologies leverages Seoul office to make Korea center of Asia-Pacific strategy. Interview with CEO Brian Hyndman, 2 March 2026. https://www.koreatimes.co.kr/economy/policy/20260302/blue-ocean-technologies-leverages-seoul-office-to-make-korea-center-of-asia-pacific-strategy. Accessed 26 April 2026

Lim BC (2026) Overnight adverse selection: Evidence from Blue Ocean ATS and NASDAQ regular trading hours. SSRN working paper 6610883. https://ssrn.com/abstract=6610883

Lin EM-H, Chen H-S, Lien D (2019) Order price clustering, size clustering, and stock price movements: Evidence from the Taiwan Stock Exchange. International Review of Financial Analysis 65:101389

Moulton PC (2005) You can't always get what you want: Trade-size clustering and quantity choice in liquidity. Journal of Financial Economics 78(1):89–119

Niederhoffer V, Osborne MFM (1966) Market making and reversal on the stock exchange. Journal of the American Statistical Association 61(316):897–916

Ohta W (2006) An analysis of intraday patterns in price clustering on the Tokyo Stock Exchange. Journal of Banking & Finance 30(3):1023–1039

Verousis T, ap Gwilym O (2013) Trade size clustering and the cost of trading at the London Stock Exchange. International Review of Financial Analysis 27:91–102

Wyckoff PR (1963) The Psychology of Stock Market Timing. Prentice-Hall, Englewood Cliffs